

Class 21: Four Important Subspaces

We can use any any $m \times n$ matrix, A , to define two important subspaces.

1.

2.

Practice problems

$$A = \begin{bmatrix} 1 & 3 & -4 & -3 & 1 \\ 0 & 1 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Find a basis for Col A .

Does Col $A = \mathbf{R}^5$?

Does Col $A = \mathbf{R}^3$?

Is the vector $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ in Col A ?

What is the dimension of Nul A ?

Is the vector $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ in Nul A ?

Find a basis for Nul A .

Definition: A linear transformation, T , from a vector space V to a vector space W is a function that assigns to each vector $\mathbf{x}$ in V a unique vector $T(\mathbf{x})$ in W , such that

This means for all vectors $\mathbf{u}$ and $\mathbf{v}$ in V and all scalars c we have

And

Any linear transformation T from V to W defines two important subspaces.

1. Range of $T =$

2. Kernel of $T =$

Key idea: If $V = \mathbf{R}^n$ and $W = \mathbf{R}^m$ and $T(\mathbf{x}) = A\mathbf{x}$ where A is some $m \times n$ matrix, then the range of T is the column space of A and the kernel of T is the null space of A .

Practice Problems #33

The set of all 2×2 matrices with real entries is a vector space denoted by $M_{2 \times 2}$. We can define a function $T : M_{2 \times 2} \rightarrow M_{2 \times 2}$ by $T(A) = A + A^T$

What is $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)$?

Prove that T is a linear transformation.

practice continued

Find one matrix in the kernel of T .

Describe all matrices in the kernel of T .

Describe all matrices in the range of T .

Find a matrix A such that $T(A) = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$